

Think-Aloud Study

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User Action	User Reaction	Designer comments	Data Encoding
Introduction to the app	"So, the software is already in the camera..."	"This app is designed for classifying terrestrial transport. It's like the app has been developed with the classifier already integrated into cameras. So, you won't just snap a pic and identify the transport. Instead, cameras you buy from us will be linked to your account with the classifier built-in."	The user understands what to expect.
Log In	"I'm looking at the app now. On the welcome page, there are text boxes for logging in. I can use my username and password to log in. It's good, there's a forgotten password message in case you forget."	The user missed the "Skip" option at the top of the Welcome page, but still managed to log in effortlessly.	User knows what to do. User does the right action.
Create Account	"I have to create an account to log in, so I tap on the Create Account button. Then, I can fill in the required details like first name, last name, email, and password. After that, I click the create account button, and I can start using the app. It opens up to the Home page."	The user missed the "Skip" option at the top of the Create Account page but managed to create an account successfully and easily. Everything was clear.	User knows what to do. User does the right action.
Explore the home page	"Here are some examples of pictures showing cars and signs specific to different transports... Hmm... Alright, let's head over to the cameras section."	The user overlooked the upper part of the Home page, solely concentrating on the Use Cases section. Confusion arose because all the Use Cases visual elements shared the same picture as an example, and there was no accompanying text in the space for related information next to each of the use cases' pictures. The absence of interactive elements on this page left the user unsure of what actions could be taken.	The user is confused. The user is unsure what to do.

Explore the previews of all the cameras of a specific environment	"I'm in the camera section, and I notice multiple cameras displaying the same shot..."	The user immediately clicked on one of the cameras without exploring the page further.	The user is unsure what to do.
Switch camera environments/locations	"Hmm..."	The user clicked on the hamburger menu, but nothing happened as it wasn't linked to any action, page, or element yet. Consequently, they were uncertain about what else they could do on the Cameras page besides clicking on one of the cameras.	The user is unsure what to do.
Open a specific camera for further analysis of the recording	"I click on Camera 1, and it takes me to the Analysis section for that camera. Here, the camera can identify the transport from the picture/recording. It predicts the vehicle, speed, timestamp... Hmm... We can see what kind of predictions it can make... However, there's no analysis filled, only the picture of the bus from the back."	The user was confused about the details the camera could identify because they were not filled in yet. They attempted to click on the details to see what would happen, but nothing occurred, leaving the user unsure about the interactiveness of this page.	The user is confused. The user is unsure what to do.
Explore the User Account	"Here, we can go to the account page where we can view cameras, history, settings, and privacy & policy... Hmmm..." Me: "Those pages are not ready yet."	The user attempted to click on all the subpages listed in the menu on the Account page, but none of them worked because they were not ready yet. This left the user confused.	The user is confused.
Log out	"Okay, and here's the option to log out. Once clicked, we're back to the Welcome page. From here, we can skip the login process and dive right into the app."	The user effortlessly clicked on the Log out button. Then he notices that there is also a "Skip" option on the welcome page which leads to the home page of the app-	User does the right action. User knows what to do.
Further feedback	U: "The features on the app are really good, everything is simple to use from making the account to the options of the cameras, but it needs to be filled with information. I just hope to see more... different kinds of transport that the cameras can identify. Here on the cameras page, I only see one example. It's a great idea that the cameras are able to identify road traffic signs. Maybe you can use it to provide information about rules that apply for each vehicle. I think the home page should clarify how the app should work and what we need to press. That would be more useful because here on the Cameras and Analysis page, I felt kind of lost. Some kind of introduction to the app would be nice." D: "Unfortunately, the cameras are not able to identify the road traffic signs, the picture with the traffic sign was only used as an example. Can I ask you some questions about the app?" U: "Yes, of course." D: "So, what would you expect these elements on the Home Page to be?"		

	U: "I thought those elements are supposed to clarify what all those pictures next to them are meant to be, making it easier for users to know what they can expect from the cameras." D: "I'll tell you what it's intended to be. So, the element in the upper part of the Home page is meant to be news or customer reviews from those who use our system. These lines next to it will be text, but it's not there yet. The use cases elements should be like images linked to short articles explaining how our cameras can be used, such as monitoring restricted areas, private land, or by emergency services." U: "Oh, so it will not be showing you a live recording from the cameras, only some pictures." D: "On the Home Page, yes." U: "So, the cameras button leads to the Cameras page where that single user can see multiple cameras that he owns..." D: "Exactly, that's why you are logged in. You're logged into your account and therefore you can see cameras that you purchased from us along with the integrated classifier in them..." U: "Okay, then I have just one suggestion. On the cameras page, every picture should have a different angle or be a different picture to avoid confusing the user." D: "Yeah, the pictures will be different, don't worry. I just put in only one picture as an example for the prototype. I wanted to ask you about this feature, the hamburger menu, because you were trying to click on it, but it didn't work. What would you expect to happen after you click on it?" U: "I thought the three stripes would lead me to a menu to select which camera I want to see, or have an option to see all the cameras or just one of them." D: "So basically, you were expecting something interconnected between the Cameras page and the Analysis page. What I intended with this is that if, for example, it's a government body or the police using it, they have cameras in different areas. By clicking on this button here, they can choose the location, and here they can see all the cameras from that location." U: "Okay, so then just add that when you press on the three stripes, you can choose the location. After selecting the location, you can choose which camera to view. It will give more options to the user. In general, I must say that the interface is very user-friendly and easy to understand."
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User Action	User Reaction	Designer comments	Data Encoding
Introduction to the app	"Okay. Clear."	"This app is designed for classifying terrestrial transport. It's like the app has been developed with the classifier already integrated into cameras. So, you won't just snap a pic and identify the transport. Instead, cameras you buy from us will be linked to your account with the classifier built-in."	The user understands what to expect.
Log In	U: "I see the welcome page with text boxes for logging in. Should I also put in my username and password?" D: "No, it is not working yet." U: "Okay, then I click on the button Log In which leads to the Home page."	The user missed the "Skip" option at the top of the Welcome page, but still managed to log in effortlessly.	User knows what to do. User does the right action.
Explore the home page	U: "I am trying to click on each of the elements on this page to see if they lead to a new page... Oh, nothing works here. What's this?" (referring to the elements on the home page with images) D: "Those elements will feature customer reviews or news. Live streaming from public cameras won't be available; instead, there will be a picture with accompanying text for users to read. Below, you'll find the use cases outlined in my creative brief. The text will highlight how the classifier with the cameras can be utilized, such as by the police for area monitoring or by the city for traffic surveillance and urban planning. Each use case will be presented individually, allowing users to explore the various applications of the cameras if they're considering a purchase. If users haven't purchased the system yet, they won't see any cameras in the cameras tab. Instead, they'll find a link leading to the website where they can make the purchase."	The user overlooked the upper part of the Home page, solely concentrating on the Use Cases section. The absence of interactive elements on this page left the user unsure of what actions could be taken.	The user is unsure what to do.
Explore the previews of all the cameras of a specific environment	U: "I am in the camera section now... so let's say I own 6 cameras... and they're all different?" D: "Yes, they are, but they are using only one picture for now as an example."	The user immediately clicked on one of the cameras without exploring the page further.	The user is confused.
Switch camera environments/locations	"Hmm... So this is nothing..."	The user clicked on the hamburger menu, but nothing happened as it wasn't linked to any action, page, or element yet. Consequently, they	The user is unsure what to do.

		were uncertain about what else they could do on the Cameras page besides clicking on one of the cameras..	
Open a specific camera for further analysis of the recording	"I click on Camera 1 and it leads me to the Analysis section for that camera. Hmm... nothing stands out here. I'm not sure if any of this analytical information is supposed to stand out, but they don't. The predicted vehicle label might be a bit bigger since it's a classifier, and placed on top."	The user was confused about the details the camera could identify because they were not filled in yet. They attempted to click on the details to see what would happen, but nothing occurred, leaving the user unsure about the interactiveness of this page.	The user is confused. The user is unsure what to do.
Explore the User Account	"I'm clicking on the round account picture at the top of the page... okay, so this leads to the same page as the account tab at the bottom of the app? That feels unnecessary because everyone understands that this round icon would be for your account. It feels redundant; it wouldn't bother me, but just to save space, I suppose."	The user understood that there are two ways to get to the account page: through the Account tab or through the round icon at the top of every page. It was clear to him.	User knows what to do.
Log out	"Okay, and here I can log out."	In the account menu, the user easily found the Log out option.	User does the right action.
Create Account	"Oh, nice. When I click on the Create Account button, I can create a new account by adding the first name, last name, email, and password. Then I can start using the app. It looks clean."	The user missed the "Skip" option at the top of the Create Account page but managed to create an account successfully and easily. Everything was clear.	User knows what to do. User does the right action.
Further feedback	D: "Can I ask you a question about my app?" U: "Yes, sure." D: "I saw you were clicking on this icon (hamburger menu) but it didn't work. What would you expect this button to be?" U: "I don't know... I saw a hamburger menu so I just thought a menu would pop out." D: "Okay, there will be a drop-down with all the locations. For example, if a city is using this app and they have cameras in different locations that they want to monitor separately, they will be able to choose the location and then see all the cameras within that location."		

User Action	User Reaction	Designer comments	Data Encoding
Introduction to the app	Okay. Clear. I'm happy to be here to give you some feedback. ”	"This app is designed for classifying terrestrial transport. It's like the app has been developed with the classifier already integrated into cameras. So, you won't just snap a pic and identify the transport. Instead, cameras you buy from us will be linked to your account with the classifier built-in."	The user understands what to expect.
Log In	"Now, I'm going to use the application that you made. So, as we can see here, we can create an account in your application, or if we already have an account, we can just log in with our username and password."	The user missed the “Skip” option at the top of the Welcome page, but still managed to log in effortlessly.	User knows what to do. User does the right action.
Create Account	“And since I do not have any accounts here, I will just click on 'Create Account.' Here, we can write our first name, last name, email, and password. Then, we can create an account and see the home page.”	The user missed the “Skip” option at the top of the Create Account page but managed to create an account successfully and easily. Everything was clear.	User knows what to do. User does the right action.
Explore the home page	"Here, I guess we can see some examples of pictures showing cars and traffic signs... Hmm... Also, we can see four buttons at the bottom of the page that indicate individual pages... Hmm..."	The user overlooked the upper part of the Home page, solely concentrating on the Use Cases section. Confusion arose because all the Use Cases visual elements shared the same picture as an example, and there was no accompanying text in the space for related information next to each of the use cases' pictures. The absence of interactive elements on this page left the user unsure of what actions could be taken.	The user is confused. The user is unsure what to do.
Explore the previews of all the cameras of a specific environment	"Here, I will click on the cameras. Here, we can see six pictures from the cameras... I'm not exactly sure if they are the same or not, but they are similar to each other."	The user immediately clicked on one of the cameras without exploring the page further.	The user is confused.
Switch camera environments/locations	-	The user did not notice the hamburger menu for the Environment.	The user is what unsure what to do.
Open a specific camera for further analysis of the recording	“Then, if I click on the analysis option, we can see a picture of a bus and the predicted information... Hmm, I'm trying to click on the information boxes but nothing works and nothing is filled	The user was confused about the details the camera could identify because they were not filled in yet. They attempted to click on the details to see what	The user is confused.

	out... I guess we can understand from here what was the speed of the bus that was driving in this area or, uh, traffic state, timestamp, plate number..."	would happen, but nothing occurred, leaving the user unsure about the interactiveness of this page.	The user is unsure what to do.
Explore the User Account	"If we click on the account section, we can see the account that we made. It has options like history, settings, privacy & policy... I'm trying to click on the options from the menu, but nothing works... but it's fine... you can add that later."	The user attempted to click on all the subpages listed in the menu on the Account page, but none of them worked because they were not ready yet. This left the user unsure.	The user is unsure what to do.
Log out	"Okay, and here's the option to log out. Once clicked, we're back to the Welcome page. From here, we can skip the login process and dive right into the app."	The user effortlessly clicked on the Log out button. Then he notices that there is also a "Skip" option on the welcome page which leads to the home page of the app.	User does the right action. User knows what to do.
Further feedback	D: "A lot of things are not there yet because I didn't have time to do it and it's just like you said, just the preview." U: "Yeah, that's fine. It's completely fine. You can add those features in the new updates for your application. So for now, I think it's really interesting that you paid attention to some specific details. For example, in your application, we can make an account just for ourselves. It's personal, and then if we see those pictures, we can read the analysis that your app can provide. And also, h we have a section called cameras, and I guess we can see... speed cameras?" D: "Yeah, the cameras also measure speed, but it's not their primary purpose. So that's mostly like if you create an account, that means that you have purchased the cameras or the system from the company, and here you have all the cameras that you purchased. So, in this case, the user purchased 6 cameras and they are using six cameras, and they can click on... Like this is kind of like a preview of all of them, of what's happening. And then if you click on any of them, it will lead you to a specific site for that camera where you can see what exactly that camera sees and what type of vehicles it identifies, and also the speed of the vehicle..." U: "Really interesting. So for this application, I guess we have to buy a subscription, right? As you said, we have to purchase the cameras. I can just say that's really good, that you paid attention to the details because your app has a lot of features, and you can just... If you just update your app, it can get better and better over some time." D: "Can I ask you a question about my app?" U: "Yes." D: "Would you expect this icon (hamburger menu) to be if you were able to click on this, for example?" U: "Yeah, I didn't see this menu before. I guess I should expect to see like a menu from here... For now, it has just only one option that shows the environment. Hmm, but maybe later... I'm not exactly sure what you can see here..." D: "I will be a drop-down menu, exactly. So, the drop-down menu will open, and it will have different types of locations that you'll have your cameras on. So, for example, if the application is used by a city and the city is monitoring different areas separately, then you will,		

	for example, have six cameras linked to one location, then four cameras to another location, and you can switch between the locations here and then see all the cameras within that location." U: "Oh, that makes sense. That's really interesting, actually. So we can, like, set, like, four cameras for a zone inside the city and, for example, like two other cameras for a zone that's like out of the city. " D: "Yeah, exactly. And then you can switch between them in here so that you don't get confused and all the cameras if they were." U: "As I said before, it has a lot of features. That you can switch on the cameras, you can switch them off, or you can click on the menu and drop-down menu and check the different zones that you want to see. Uh, for now, I'm... I like your app and it's really interesting."
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Conclusion

In conclusion, all users were able to smoothly and effortlessly navigate through the welcome page, either by logging in or creating an account. However, some users missed the skip option, indicating a need for improved visibility. The homepage posed the most significant challenges, with users finding the use case elements confusing due to identical images and lack of accompanying text. Additionally, users overlooked the upper part of the homepage, further contributing to confusion. Most users also encountered difficulties with non-interactive elements and were unsure of their functionality.

On the camera page, users understood they were viewing previews from multiple cameras but were uncertain about further actions. Many attempted to interact with the inactive hamburger menu, expecting location options, highlighting a need for interactivity. In the analysis section, users found the details unclear and non-interactive, suggesting the need for clearer presentation and interactive features.

In the user account section, users attempted to click on non-functional menu options, indicating readiness for additional features. However, all users found the logout option easily.

In summary, my three main conclusions from the study are:

- **Highlighting Key Elements:** Certain elements of the app need better visibility, such as the "Skip" option on the welcome page and the "Predictive Vehicle" information in the Analysis section. These enhancements would ensure users can easily locate and understand crucial features.
- **Clarifying Use Case Elements:** Users struggled with the lack of clarity on the homepage and camera pages, where identical images and absence of text caused confusion. Improving the presentation of use case elements with clear text and distinct visuals is essential for enhancing user comprehension.
- **Enhancing Interactivity:** Many users attempted to interact with non-functional elements, such as the hamburger menu on the camera page, indicating a desire for interactivity. Transforming these elements into interactive features, such as dropdown menus for camera locations, would improve user experience and navigation.

Implementing these key conclusions will address usability issues and enhance overall user satisfaction with the application.